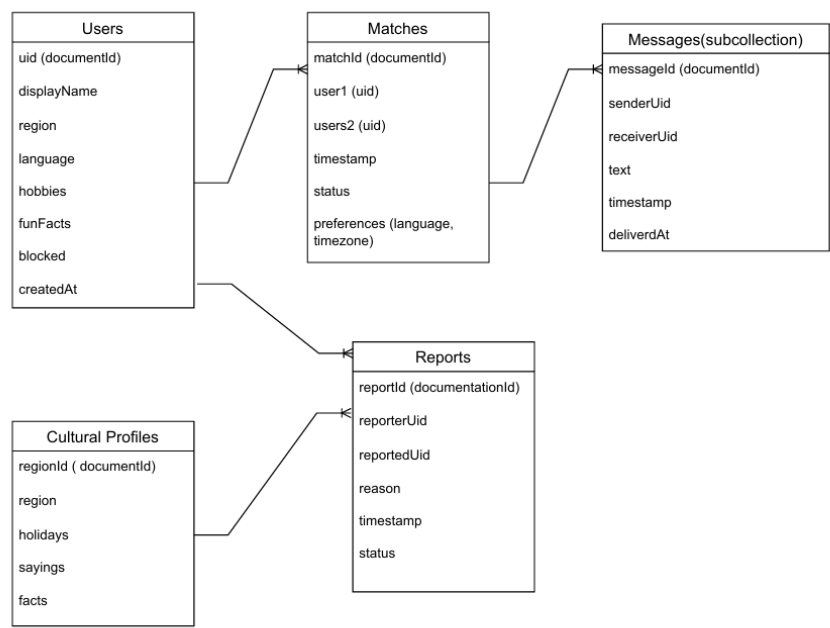


DATABASE SCHEMA

THE ER DIAGRAM



1. Entity Descriptions

Users

Stores the core details of each pen pal, including personal info, language preferences, hobbies, and blocked users. This entity is the foundation for all matchmaking and messaging operations.

Matches

Represents a pairing between two users for correspondence. Includes match metadata such as timestamps, status, and preferences.

Messages

Stores the conversation between matched users. Each message references the sender, recipient, and time sent.

Reports

Keeps track of user-reported issues such as harassment or inappropriate content, linking the reporting user, the reported user, and the context.

CulturalProfiles

Holds optional cultural information about users (hobbies, cultural background, languages spoken) used to improve match recommendations.

2. Attributes List

Users Collection

Field	Type	Constraints	Description
uid	string	Primary Key, Unique	Unique identifier for each user
displayName	string	Optional	User's display name
email	string	Required, Unique	User's email for login
languages	array of strings	Optional	Languages the user speaks
hobbies	array of strings	Optional	User interests
blockedUsers	array of strings	Optional	List of users blocked by this user
createdAt	timestamp	Default: now	Account creation time

Matches Collection

Field	Type	Constraints	Description
matchId	string	Primary Key	Unique ID for each match
user1Id	string	Required	UID of first user
user2Id	string	Required	UID of second user
status	string	Optional	e.g., active, completed
matchedAt	timestamp	Default: now	Time match was created

Messages Collection

Field	Type	Constraints	Description
messageId	string	Primary Key	Unique ID for each message
matchId	string	Required	Reference to the match
senderId	string	Required	UID of sender
recipientId	string	Required	UID of recipient
text	string	Required	Message content
sentAt	timestamp	Default: now	Time message was sent

Reports Collection

Field	Type	Constraints	Description
reportId	string	Primary Key	Unique ID for the report
reportingUserId	string	Required	UID of reporting user
reportedUserId	string	Required	UID of reported user
reason	string	Required	Description of issue
createdAt	timestamp	Default: now	Time report was filed

CulturalProfiles Collection

Field	Type	Constraints	Description
profileId	string	Primary Key	Unique ID for the cultural profile
userId	string	Required	UID of the user
languages	array of strings	Optional	Languages spoken
hobbies	array of strings	Optional	Hobbies and interests
culturalNotes	string	Optional	Any additional cultural info

3. Relationships & Cardinalities

- **Users and Matches:**

One User can participate in many Matches.

Each Match connects exactly two Users.

Cardinality: One-to-Many (User and Matches)

- **Matches and Messages:**

One Match can have many Messages.

Each Message belongs to exactly one Match.

Cardinality: One-to-Many (Match and Messages)

- **Users and Reports:**

One User can file multiple Reports.

Each Report references one reporting user and one reported user.

Cardinality: One-to-Many (User and Reports)

- **Users and CulturalProfiles:**

One User can have one optional CulturalProfile.

Cardinality: One-to-One (User and CulturalProfile)

4. Indexes & Keys

- uid serves as the unique identifier for Users.
- matchId, messageId, reportId, profileId serve as document IDs for other collections.
- Composite indexes may be added for queries such as:
 - “Retrieve all matches for a user” (user1Id or user2Id)
 - “Retrieve messages for a match” (matchId + sentAt)

5. Deployment Information

- The database is deployed on **Firebase Firestore**, a NoSQL, document-based database.
- Collections are created dynamically when data is added.
- Backend access is secured via Firebase Admin SDK and service account credentials.
- Real-time updates allow chat messages and matches to sync instantly between users.

6. Choice Justification

- **Firestore Advantages for this Project:**
 - **Real-time syncing** – essential for messaging functionality.
 - **Flexible schema** – allows optional fields like hobbies and cultural notes without strict table constraints.
 - **Seamless integration** – works smoothly with Firebase Authentication and Cloud Functions.
 - **Scalability** – handles growing number of users and matches efficiently.